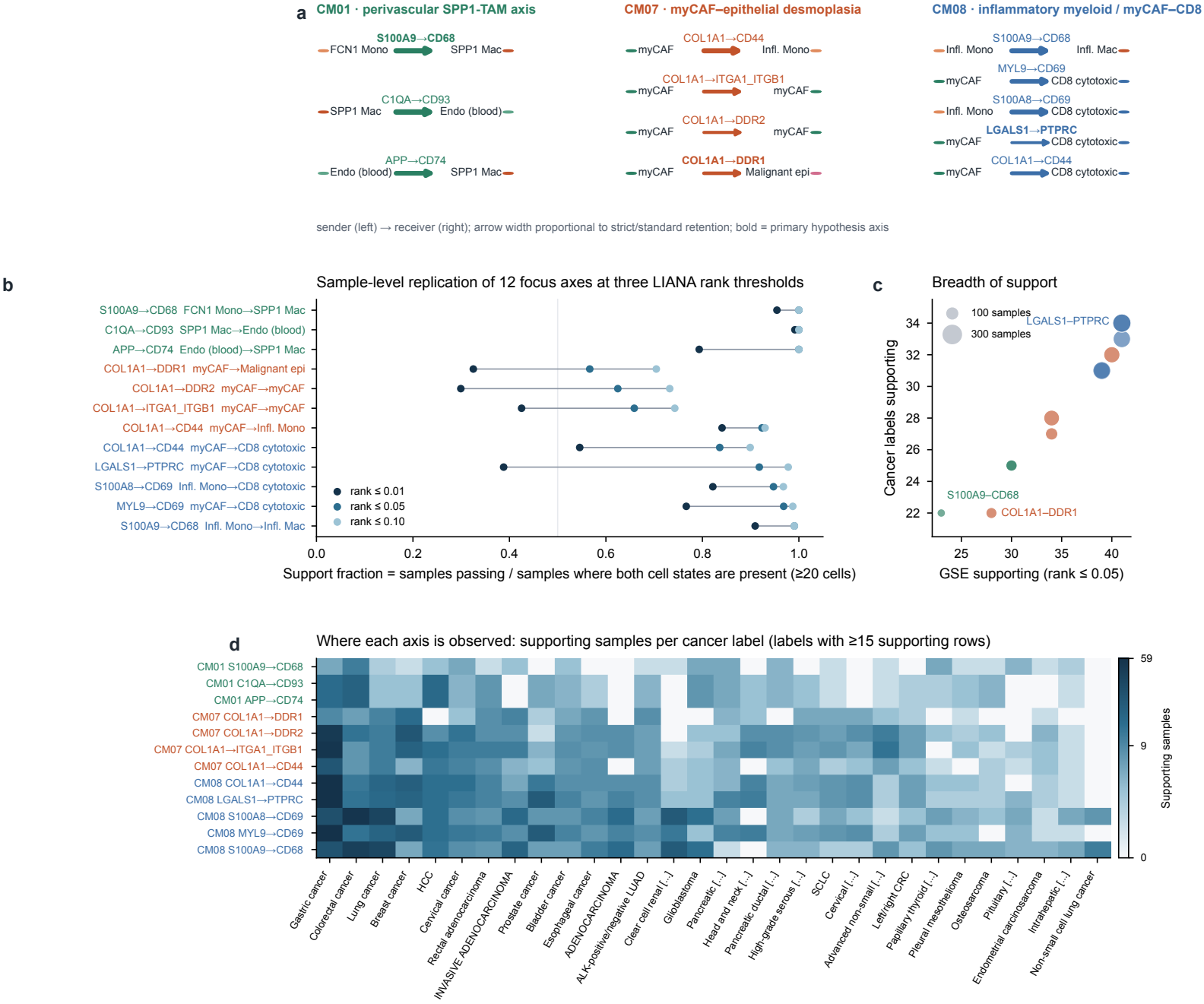

a
CM01 · perivascular SPP1-TAM axis
S100A9→CD68
FCN1 Mono
SPP1 Mac
C1QA→CD93
SPP1 Mac
Endo (blood)
APP→CD74
Endo (blood)
SPP1 Mac
CM07 · myCAF–epithelial desmoplasia
COL1A1→CD44
myCAF
Infl. Mono
COL1A1→ITGA1_ITGB1
myCAF
myCAF
COL1A1→DDR2
myCAF
myCAF
COL1A1→DDR1
myCAF
Malignant epi
CM08 · inflammatory myeloid / myCAF–CD8
S100A9→CD68
Infl. Mono
Infl. Mac
MYL9→CD69
myCAF
CD8 cytotoxic
S100A8→CD69
Infl. Mono
CD8 cytotoxic
LGALS1→PTPRC
myCAF
CD8 cytotoxic
COL1A1→CD44
myCAF
CD8 cytotoxic
sender (left) → receiver (right); arrow width proportional to strict/standard retention; bold = primary hypothesis axis
b
Sample-level replication of 12 focus axes at three LIANA rank thresholds
S100A9→CD68 FCN1 Mono→SPP1 Mac
C1QA→CD93 SPP1 Mac→Endo (blood)
APP→CD74 Endo (blood)→SPP1 Mac
COL1A1→DDR1 myCAF→Malignant epi
COL1A1→DDR2 myCAF→myCAF
COL1A1→ITGA1_ITGB1 myCAF→myCAF
COL1A1→CD44 myCAF→Infl. Mono
COL1A1→CD44 myCAF→CD8 cytotoxic
LGALS1→PTPRC myCAF→CD8 cytotoxic
S100A8→CD69 Infl. Mono→CD8 cytotoxic
MYL9→CD69 myCAF→CD8 cytotoxic
S100A9→CD68 Infl. Mono→Infl. Mac
rank ≤ 0.01
rank ≤ 0.05
rank ≤ 0.10
0.0
0.2
0.4
0.6
0.8
1.0
Support fraction = samples passing / samples where both cell states are present (≥20 cells)
c
Breadth of support
100 samples
300 samples
LGALS1–PTPRC
S100A9–CD68
COL1A1–DDR1
Cancer labels supporting
GSE supporting (rank ≤ 0.05)
22
24
26
28
30
32
34
25
30
35
40
d
Where each axis is observed: supporting samples per cancer label (labels with ≥15 supporting rows)
CM01 S100A9→CD68
CM01 C1QA→CD93
CM01 APP→CD74
CM07 COL1A1→DDR1
CM07 COL1A1→DDR2
CM07 COL1A1→ITGA1_ITGB1
CM07 COL1A1→CD44
CM08 COL1A1→CD44
CM08 LGALS1→PTPRC
CM08 S100A8→CD69
CM08 MYL9→CD69
CM08 S100A9→CD68
Gastric cancer
Colorectal cancer
Lung cancer
Breast cancer
HCC
Cervical cancer
Rectal adenocarcinoma
INVASIVE ADENOCARCINOMA
Prostate cancer
Bladder cancer
Esophageal cancer
ADENOCARCINOMA
ALK-positive/negative LUAD
Clear cell renal [...]
Glioblastoma
Pancreatic [...]
Head and neck [...]
Pancreatic ductal [...]
High-grade serous [...]
SCLC
Cervical [...]
Advanced non-small [...]
Left/right CRC
Papillary thyroid [...]
Pleural mesothelioma
Osteosarcoma
Pituitary [...]
Endometrial carcinosarcoma
Intrahepatic [...]
Non-small cell lung cancer
Supporting samples
59
9
0